

Safety-Aware Aggregation for Federated Fine-Tuning of LLMs

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Abstract. The rapid democratization of Large Language Models (LLMs) has amplified a dual-use dilemma: the same capabilities that let these models advance legitimate science also let them generate unsafe content in domains such as biosecurity and chemistry. At the same time, the demand for privacy-preserving training has driven the adoption of Federated Learning (FL), in which a central server aggregates client updates but never sees the raw local data. This privacy guarantee disables the dominant safety paradigm: server-side inspection and filtering of training data becomes impossible, so a client fine-tuning on hazardous data can silently raise the hazardous capability of the global model. Our key observation is that, although the server cannot select data, it can still act on a standardized risk signal that each client computes locally and shares without ever exposing and dropping raw samples. The goal of this work is therefore to relocate safety enforcement from the data space into the aggregation space, using this risk signal to properly weight the incoming updates. Building on a Parameter-Efficient Fine-Tuning (PEFT) approach, we evaluate a spectrum of aggregation strategies, including a novel Federated Risk-Aware Projection (RAP-FL). Across three client data distributions, we quantify the trade-off between suppressing hazardous knowledge (WMDP) and preserving general-domain utility (MMLU), and find that RAP-FL gives the most favorable safety-utility balance among the strategies we tested.

Keywords: Federated Learning · AI Safety · Large Language Models · PEFT · Trustworthy AI

1 Introduction

As Large Language Models (LLMs) are deployed in settings where their decisions, or generated outputs, have real-world consequences, ensuring their safe behavior has become a central requirement for trustworthy AI. This is the goal of *alignment*: the process of shaping model behavior so that its outputs reflect human intentions and values, rejecting harmful requests rather than complying with them [15]. An unaligned model can actively cause harm, making the gap between raw capability and intended behavior a safety problem rather than merely a quality one. The stakes are highest in *dual-use* settings, where the

same knowledge is benign in one context yet dangerous in another: explaining how a pathogen spreads can help a doctor stop an outbreak or help an attacker start one. This deserves particular attention because such harm is severe and irreversible, and because the dangerous knowledge is hardly separable from legitimate scientific expertise [16]. Unsafe content arises in the first place because pre-training ingests vast, largely uncurated corpora: a base model learns to predict plausible text, with no built-in preference for safety, and therefore absorbs hazardous information alongside useful knowledge. Fine-tuning is an effective remedy because it adjusts model behaviour toward aligned, safe responses without retraining from scratch [15]. Until recently, this remained costly, but Parameter-Efficient Fine-Tuning and Low-Rank Adaptation (LoRA) in particular [8] have reduced the computational overhead by orders of magnitude, making adaptation feasible even on modest hardware. This computational affordability aligns well with the rise of *Federated Learning* (FL) [13], a paradigm in which many clients collaboratively train a shared model while their raw data never leave their devices, and only weight updates are sent to a central server. Fine-tuning LLMs in such a setting is attractive for privacy-sensitive domains [6], but it introduces the *blind server problem*: because the server never sees the local data, it cannot directly filter or select what each client is teaching the model. A client that fine-tunes on hazardous data therefore injects that capability into the global model, which the server aggregates without any way of noticing it. The same blindness is what makes deliberate poisoning possible [17], but the failure mode we study does not require a hostile participant: it is enough that one member of the federation holds legitimate data whose content is dual-use. The central question of this work is how to preserve safety when the server cannot inspect what it aggregates. To address it, we explore several risk-aware aggregation strategies that move safety enforcement from the data space into the aggregation space, where the server is permitted to operate. The key idea is to estimate the reliability of each client data and pass this signal to the server, which uses it to weight, steer, or geometrically reshape the incoming updates. Our evaluation across realistic data-distribution scenarios shows that the most robust of these strategies achieves a strong safety-utility trade-off, suppressing hazardous knowledge without degrading general capability.

2 Related Work

Ensuring that LLMs behave safely has motivated a substantial body of work on measuring and mitigating their dual-use risks. The retention of hazardous knowledge is now systematically quantified by benchmarks such as WMDP [12], which probes capabilities across biosecurity, cybersecurity, and chemical domains and provides the threat metric we adopt. To remove such capabilities once a model has acquired them, the community has largely pursued post-training interventions: machine unlearning methods such as RMU [22] perturb internal representations to suppress hazardous knowledge, Task Vector [9] operates directly in weight space by negating the task vectors associated with dangerous

skills, and complementary circuit-breaking techniques [23] interrupt harmful generative pathways through representation engineering. These approaches are reactive by design: they presuppose that the unsafe capability already exists in the model. A more tamper-resistant philosophy instead prevents acquisition at the source. The Deep Ignorance framework [14] relies on the principle that the strongest safeguard is to ensure the model never encounters hazardous information during training. It applies a multi-stage data-filtering pipeline in which a keyword blocklist removes documents containing domain-specific hazardous terms, while flagged documents are escalated to a fine-tuned ModernBERT [21], a Masked Language Model classifier that scores their threat probability, discarding data above a strict threshold. Models trained on the resulting sanitized corpora strongly resist adversarial fine-tuning aimed at reviving suppressed capabilities, at minimal cost to utility. Deep Ignorance is the primary safety baseline of our study, yet its core mechanism, based on direct inspection and filtering of training data, is precisely what becomes arguable once training is decentralized because the server entity cannot directly control it. Recently, decentralized fine-tuning has become a realistic and feasible scenario, thanks to the application of efficient adaptation methods and the federated paradigm itself. Realizing fine-tuning at scale has become practical with Parameter-Efficient Fine-Tuning. LoRA [8] injects low-rank trainable matrices into otherwise frozen transformer layers, reducing the number of trainable parameters by several orders of magnitude. In addition, model quantization reduces a network’s memory footprint by using lower-precision weights, and QLoRA [5] extends this approach by quantizing the frozen base model to 4-bit precision, enabling efficient adaptation on consumer-grade hardware with minimal performance degradation [10]. These efficiency gains have made it feasible to fine-tune LLMs within FL [13], enabling collaborative training without centralizing raw data and making it well suited to privacy-sensitive domains. FedAvg [13] remains the dominant aggregation rule, computing a data-weighted average of client updates, and recent FedPEFT frameworks [19, 11] integrate LoRA into this pipeline to cut communication cost down to a few lightweight adapter matrices (FedLoRA [18]). This decentralization, however, is what makes safety hard to enforce: the server observes only weight updates and controls only the aggregation mechanism, not the data. The server therefore remains semantically blind to the knowledge encoded in the incoming adapters, and no existing FedPEFT method reconciles this privacy constraint with the data inspection that the safety paradigm demands. Filtering can still be delegated to the client, which does hold the data, but whether this is the best use of a risk signal in a federation is an empirical question, and the one this work sets out to answer. Our work targets this gap, with risk-aware aggregation strategies that enforce safety inside the LoRA weight space instead of selecting data. They require a risk signal that each client computes locally and reports honestly, an assumption that is reasonable in a cross-silo federation of a few identifiable institutions, bound by an agreement and subject to audit. The participants are assumed honest: the concern is not a saboteur, but the fact that legitimate scientific data is densely dual-use.

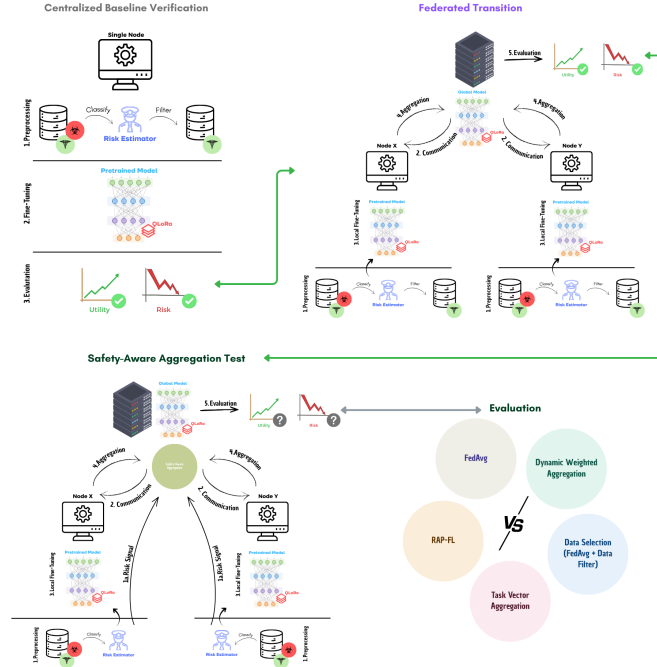


Fig. 1. Methodology Overview

3 Methodology

Our goal is to preserve source-level safety guarantees within a FL pipeline through dedicated aggregation mechanisms for local client updates, driven by a prior data-reliability estimate. Aggregation is the only FL stage where the server is allowed to act against unsafe behaviour, and in our setting it operates on the LoRA adapters tuned locally by each client. We organize the work as a four-module pipeline (Fig. 1), where each module is a self-contained experiment whose outcome gates the transition to the next one. The four modules are: (i) a *Centralized Baseline Verification*, which checks that data-filtering safety survives a parameter-efficient regime; (ii) a *Federated Transition*, which re-establishes the same behaviour under a Federated paradigm and fixes the reference baselines; (iii) *Safety-Aware Aggregation Test* phase implements and runs FL configurations with safety-aware aggregation mechanisms; and (iv) an *Evaluation* module, which compares all strategies applied on the safety-utility trade-off.

3.1 Centralized Baseline Verification

The first module reproduces the Deep Ignorance safeguard [14] on a single node using a low-rank, 4-bit-quantized adapter (QLoRA). Since Deep Ignorance was

validated with full-parameter training, this bottleneck could plausibly alter the dynamics that make data filtering effective; we therefore verify the mechanism in a centralized setting before incurring the cost of federation. As shown in Fig. 1, the module runs three steps. In **(1) Preprocessing**, a *Risk Estimator*, a fine-tuned ModernBERT classifier [21], scores every document and filters out the hazardous ones, yielding a sanitized dataset. In **(2) Fine-Tuning**, the frozen pre-trained model is adapted on controlled dataset compositions through QLoRA. In **(3) Evaluation**, the adapted model is measured on Utility and Risk benchmarks. The purpose is to confirm two preconditions: that exposure to hazardous data measurably raises hazardous capability, and that excluding it suppresses that capability at an acceptable utility cost. Only when this pattern holds, Risk decreasing while Utility is preserved (the green-check outcome in Fig. 1), the safeguard is considered transposable and the pipeline moves to the Federated setting.

3.2 Federated Transition

The second module applies the previous approach under FL. The frozen pre-trained base model θ_0 is distributed to all clients; only the lightweight LoRA adapters ϕ are trained and exchanged.

Each client, therefore, performs the federated counterpart of the centralized steps (Fig. 1): **(1) Preprocessing**, where the local Risk Estimator classifies and filters the data; **(2) Communication**, which exchanges adapters between clients and server; **(3) Local Fine-Tuning** of the QLoRA adapters; and **(4) Aggregation**, where the server combines the updates. The aggregation rule used at this step is FedAvg, a data-weighted average of the client adapters:

$$\Delta\Phi_{global} = \sum_{k=1}^K \frac{n_k}{N} \Delta\Phi_k \quad (1)$$

where $\Delta\Phi_k$ is the adapter update computed by client k , weighted by its share of local training samples n_k/N , with N the total number of samples across all clients. This module fixes the two references that propagate to the final comparison: *standard FedAvg* with no intervention, which establishes the upper bound for vulnerability, and *FedAvg on locally filtered data* (Data Selection), the direct federated transposition of Deep Ignorance [14]. A concluding **(5) Evaluation** verifies that the safety pattern, Risk down and Utility preserved, still survives once training is decentralized; only a positive outcome justifies experimenting with the safety-aware aggregation mechanisms of the next module.

3.3 Safety-Aware Aggregation Test

This module is the core contribution. It keeps the FedLoRA backbone but removes data filtering. Each client now emits only a *Risk Signal* (step 1a in Fig. 1): the per-sample threat probability $P_{threat}(x)$ produced by the shared Risk Estimator and sent alongside the adapter update. The server consumes this signal

inside a *Safety-Aware Aggregation Mechanism*. We investigate three candidate solutions and we evaluate their trade-offs empirically in Section. 5.

M1: Dynamic Weighted Aggregation. The Risk Estimator determines the proportion of safe and hazardous samples associated with a client. Safe samples contribute a score of 0.8, while hazardous samples contribute a score of 0.2 to the aggregation process. For a client k processing $N_{S,k}$ safe samples and $N_{T,k}$ hazardous samples, the risk score is defined as follows:

$$w_k = \frac{0.8 \cdot N_{S,k} + 0.2 \cdot N_{T,k}}{N_{tot,k}} \quad (2)$$

The global update is then computed *without* renormalization:

$$\Delta\Phi_{global} = \sum_{k=1}^K w_k \Delta\Phi_k \quad (3)$$

We deliberately omit normalization so that the magnitude of the global update directly reflects the underlying data reliability: clients holding hazardous data contribute proportionally less, shrinking their influence on the model instead of being hard-excluded.

M2: Task Vector Aggregation. Inspired by Ilharco et al. [9], this method isolates hazardous capabilities into a dedicated task vector. Each client independently trains two LoRA adapters on its data partitioned by the Risk Estimator: a safe adapter $\Phi_{safe,k}$ and a threat adapter $\Phi_{threat,k}$. The server aggregates these via FedAvg into global safe and threat adapters, then applies vector negation scaled by a factor α :

$$\Phi_{t+1} = \Phi_{safe_global} - \alpha (\Phi_{threat_global} - \Phi_t) \quad (4)$$

Because federated training is iterative, subtracting Φ_t is mathematically essential to prevent exponential forgetting: it ensures the negation targets only the newly acquired hazardous knowledge $\Delta\Phi_{threat_global}$, fully preserving the model’s historical utility across rounds.

M3: Federated Risk-Aware Projection (RAP-FL). M1 and M2 reduce the influence of a client’s whole update or completely remove an update that seems risky. However, harmful and useful information are often mixed together in the same update (dual-use problem). As a result, removing or weakening the entire update can also throw away valuable knowledge. RAP-FL takes a different approach. Instead of treating the whole update as dangerous, it identifies the specific direction in the model’s parameter space where harmful knowledge is being learned. It then removes only that harmful part while keeping the rest of the update. This allows the model to retain useful knowledge and reduce the loss of beneficial information.

Each client uses its local threat score $P_{threat}(x)$ as a soft mask to produce two updates, a utility update $\Delta\Phi_k^U$ (low-risk samples) and a risk update $\Delta\Phi_k^R$ (high-risk ones). The server averages them, weighting each client by its dataset size a_k , into a useful direction G^U and a hazardous direction G^R , and projects every update off G^R via the Frobenius inner product $\langle \cdot, \cdot \rangle_F$:

$$\text{Proj}_{G^R}(V) = \frac{\langle V, G^R \rangle_F}{\|G^R\|_F^2} G^R. \quad (5)$$

Subtracting this component (scaled by $\lambda, \mu \in [0, 1]$) gives the cleaned updates $\tilde{\Delta}\Phi_k^U, \tilde{\Delta}\Phi_k^R$. Because the risk update may still hold benign knowledge dragged in by dual-use samples, the server keeps the residual R_k of $\tilde{\Delta}\Phi_k^R$ aligned with G^U rather than discarding it, and forms the global model as

$$\Phi_{t+1} = \Phi_t + \sum_k a_k \tilde{\Delta}\Phi_k^U + \gamma \sum_k a_k R_k, \quad (6)$$

where λ, μ control how hard hazardous directions are erased and $\gamma \in [0, 1]$ how much salvaged utility is restored.

3.4 Evaluation

The final module evaluates the safety-aware methods by comparing every strategy head-to-head on the safety-utility trade-off. We evaluate each method by probing its **latent knowledge** [2], the factual content encoded in the weights, through a normalized Multiple-Choice Question Answering (MCQA) framework.

In MCQA, the model is presented with a question and a fixed set of candidate answers, and must select one. It does not generate free text: instead, each candidate is scored directly from the model’s output probabilities [1], and the candidate the model finds most likely is taken as its answer. To compare candidates of different lengths fairly, a candidate’s score is the *average* log-probability of its tokens (length normalization), so longer answers are not penalized. Formally, for a prompt Q of M tokens and candidates $\{C_1, \dots, C_k\}$, the predicted answer $\hat{y}$ is the highest-scoring (with N the full sequence length and $L = N - M$ the answer length):

$$\hat{y} = \arg \max_{j \in \{1, \dots, k\}} \frac{1}{L} \sum_{i=M+1}^N \log P(t_i | t_1, \dots, t_{i-1})$$

The reported metric is then simply **accuracy**: the fraction of questions answered correctly, $N_{\text{correct}}/N_{\text{total}}$.

We evaluate across two aspects: **Threat** (sensitive/hazardous knowledge) and **Utility** (general-purpose benign knowledge). The efficacy of an intervention is determined by its ability to optimize the safety-utility trade-off: a successful method must minimize accuracy on the Threat target while preserving relatively high accuracy across the general Utility benchmarks.

4 Experimental Setup

4.1 Models

All experiments are based on **deep-ignorance-pretraining-stage-unfiltered**¹ [14], a 6.9B-parameter decoder-only model trained on 500B unfiltered tokens, an ideal baseline as it encodes dual-use knowledge prior to any post-training alignment. For client-side risk assessment, we deploy a fine-tuned **ModernBERT-large** classifier [21], inherited from the Deep Ignorance pipeline, which assigns a continuous threat probability $P_{threat}(x) \in [0, 1]$ to each training sample.

4.2 Datasets

Training data consists of **Camel Chemistry** [4] and **Camel Biology** [3], two dense, dual-use scientific corpora that were also used in the original Deep Ignorance study, ensuring strict experimental consistency. After formatting and labeling the data through the Risk Estimator (threshold $\tau = 0.8$, above which samples are labeled *unsafe* and the rest *safe*), we construct three fixed-composition datasets of 13,314 samples each: a *Safe* dataset (100% safe samples), a *Mixed* dataset (50% safe and 50% threat samples), and a *Threat* dataset (100% threat samples).

For the evaluation phase, we employ two standardized MCQA benchmarks:

- **WMDP** [12] for measuring hazardous knowledge retention (*Threat*).
- **MMLU** [7] [20] is used to measure both benign domain knowledge, *MMLU-Bio* and general knowledge *MMLU-NoBio* (*Utility*).

4.3 Experimental Setup and Hyperparameters

To ensure computational feasibility and a strictly controlled environment, all experiments utilize a unified Quantized Low-Rank Adaptation (QLoRA) regime on the 6.9B base model. The base weights are frozen and quantized to 4-bit NormalFloat (NF4) with double quantization, while activation gradients use 16-bit precision (`float16`).

LoRA adapters are injected into all linear layers (attention and MLP blocks), restricting the trainable parameter space to exactly 33.5M parameters ($\sim 0.49\%$ of the network). Optimization is driven by `paged_adamw_8bit` with an effective global batch size of 32 and a cosine learning rate scheduler. The complete architectural and optimization configurations used are reported inside the repository² for detailed analysis and reproducibility purposes.

¹ <https://huggingface.co/EleutherAI/deep-ignorance-pretraining-stage-unfiltered>

² https://anonymous.4open.science/r/Safety-Aware_Aggregation_for_Federated_Fine-Tuning_of_LLMs

4.4 Federated Simulation

We simulate a $K=2$ client network, chosen to isolate the algorithmic performance of our safety mechanisms from data heterogeneity effects. Each client holds 6,016 training samples; a global validation set of 1,282 samples is held out at the server. Synchronization and aggregation occurs every 47 local steps ($1/4$ of a local epoch).

To stress-test our methods under varying network threat levels, we define three experimental scenarios reported in Table 1.

Scenario	Client	Safe Samples	Threat Samples	Total
S1: Extreme Asymmetry	C1	6,016 (100%)	0 (0%)	6,016
	C2	0 (0%)	6,016 (100%)	6,016
S2: Mixed Homogeneous	C1	3,008 (50%)	3,008 (50%)	6,016
	C2	3,008 (50%)	3,008 (50%)	6,016
S3: Mixed Heterogeneous	C1	3,609 (60%)	2,407 (40%)	6,016
	C2	1,804 (30%)	4,212 (70%)	6,016

Table 1. Safe and Threat data distribution per client across the three federated scenarios.

5 Results and Discussion

5.1 Centralized Baselines

Figure 2 (Left) illustrates the models’ performance on the WMDP benchmark, our primary proxy for hazardous knowledge acquisition. Continuous fine-tuning on the Threat-Only dataset causes a clear escalation in malicious capabilities, driving accuracy from the base model’s 33.62% to a peak of 41.32% (at step 470). Crucially, the Mixed Baseline exhibits a similarly dangerous trajectory, climbing to 40.06%. Conversely, the Safe-Only baseline successfully drops malicious context, keeping WMDP accuracy suppressed and ending slightly below its pre-training levels (33.23% at step 1880). Simultaneously, we observe the impact of these data compositions on general domain utility, measured via the MMLU-Bio benchmark (Figure 2 (Right)). The analysis reveals that the Threat-Only baseline not only maximizes malicious capabilities but also significantly improves general biological utility, increasing MMLU-Bio accuracy from the baseline of 36.34% to a peak of 44.83% due to the information enrichment. The Mixed baseline follows a trend similar to Threat throughout the five epochs, while the Safe baseline initially improves utility before declining sharply. The Safe-Only baseline maintains strong utility during the first three epochs (40-42% accuracy), but extended training results in a noticeable drop, reaching 34.58% accuracy in the final epoch. We attribute this degradation to the filtering process, which removes diverse scientific content. Although it is enough for safe short-term adaptation, the limited variety of data makes the model more likely to overfit during extended fine-tuning, which can reduce its overall reasoning ability.

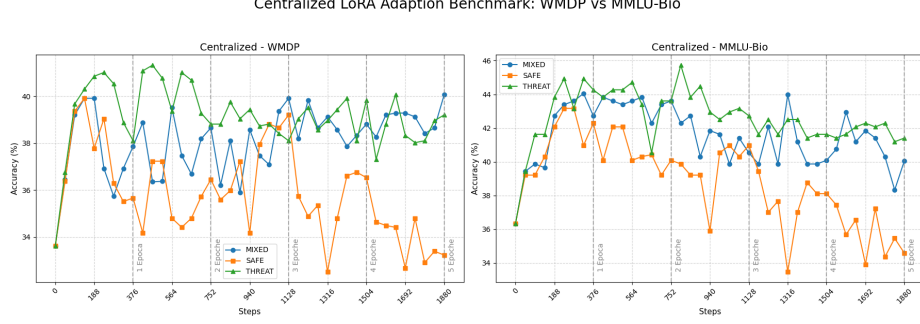


Fig. 2. Centralized PEFT baselines evaluation. **Left:** WMDP accuracy demonstrates rapid hazardous capability acquisition for Threat and Mixed datasets, while the Safe-Only baseline remains successfully suppressed. **Right:** MMLU-Bio accuracy highlights the eventual utility degradation of the Safe-Only baseline due to variance loss, whereas Threat-Only improves general scientific utility.

5.2 Federated Solutions Results

To evaluate the proposed aggregation algorithms, we analyze their performance in the three data distribution scenarios (Extreme Asymmetry, Mixed Homogeneous and Heterogeneous). The evaluation relies on six comparative plots that display WMDP (Threat) and MMLU-Bio (Utility) metrics across 940 local steps (5 epochs).

Standard Baselines. The standard FedAvg algorithm (Method 0) fails to maintain safety. In all scenarios, the aggregation of malicious updates causes a rapid escalation in threat capabilities, with WMDP accuracy peaking between 40% and 41.5%. While this preserves excellent general utility (MMLU-Bio > 42%), the model fails in its safety objective. Applying the *Deep Ignorance* filtering logic at the edge (Method 1) reveals significant limitations in decentralized environments. In the Extreme Asymmetry scenario (S1), Method 1 completely excludes the compromised client, successfully keeping WMDP around 33%. However, this drastic data reduction triggers the same variance-loss overfitting observed in the centralized baselines, dropping MMLU-Bio accuracy to 37.0%. More critically, in the Heterogeneity scenario (S3), Method 1 fails on both fronts: WMDP climbs to nearly 40% while MMLU-Bio degrades to 38.5%.

The Limits of Dynamic Weight and Task Vector Strategies. Method 2 (Dynamic Weighted Aggregation) exhibits extreme variance depending on the network topology. In S1, where local batches are purely safe or purely hazardous, the dynamic scalars perfectly penalize the rogue node, maintaining WMDP at 33.5% while preserving outstanding utility (MMLU-Bio ~43%). However, when safe and threat samples are entangled within the same local datasets (S2 and S3), the batch-level scores average out. Consequently, the dampening effect vanishes, and Method 2 completely collapses, allowing WMDP to soar above 43%

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Benchmark Results: WMDP vs MMLU-Bio (All Scenarios)

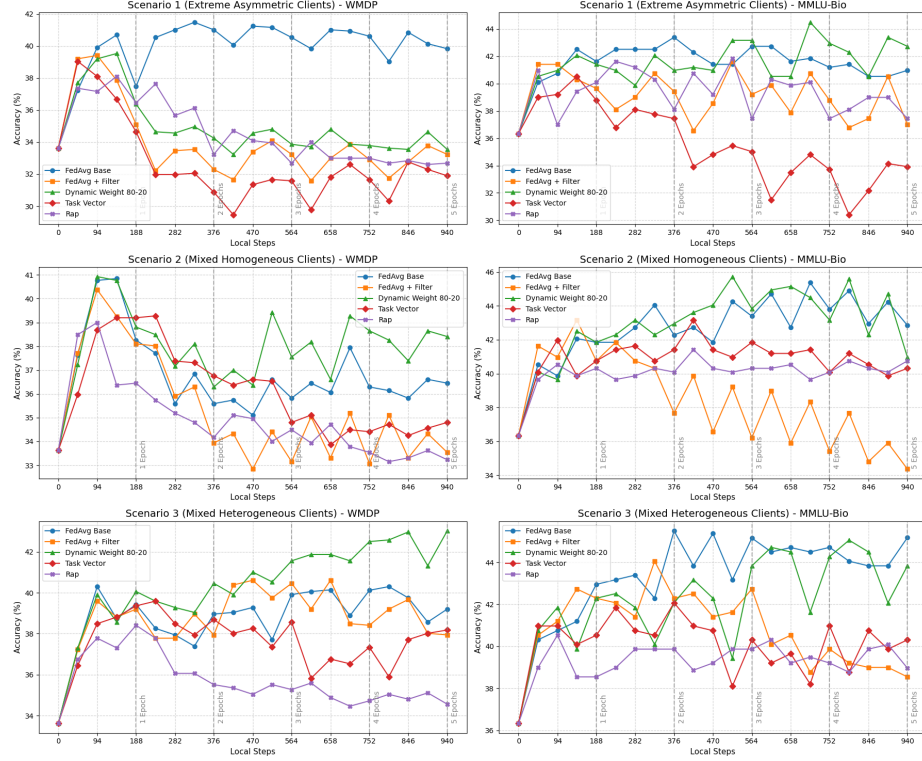


Fig. 3. WMDP and MMLU-Bio benchmark accuracy across the three federated scenarios throughout five training epochs.

in S3. Method 3 (Federated Task Vector Aggregation) proves to be a highly aggressive intervention. In S1, the isolated threat vector is so potent that its negation induces severe catastrophic forgetting, driving both WMDP (31.8%) and MMLU-Bio (33.9%) below pre-training levels. In the Realistic scenario (S3), the mathematical extraction of the task vector becomes noisy due to variance in distribution, struggling to fully suppress the threat (WMDP $\sim 38.2\%$).

Risk-Aware Projection (RAP-FL). Method 4 (RAP-FL) emerges as the strictly superior and most robust algorithm. Thanks to its per-step geometric projection and utility salvage mechanisms, RAP-FL proves to be entirely *scenario-agnostic*. Across S1, S2, and S3, it consistently neutralizes hazardous trajectories before they consolidate. In the highly challenging realistic scenario (S3), RAP-FL successfully suppresses WMDP accuracy to 34.5% (safely below the baseline) while maintaining robust general utility at 39.0% (well above the baseline). By geometrically decoupling benign concepts from malicious parameter sub-spaces,

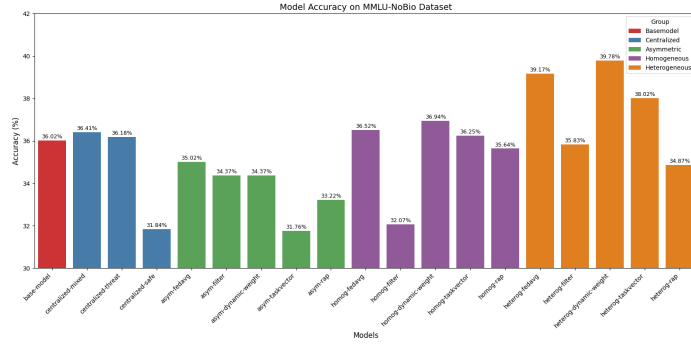


Fig. 4. Catastrophic Forgetting Evaluation on General Knowledge (MMLU-NoBio).

RAP-FL uniquely achieves proactive safety in decentralized networks without inducing catastrophic forgetting or relying on homogeneous data distributions.

General Knowledge Benchmarks. To verify that safety interventions do not degrade broader reasoning capabilities beyond the targeted scientific domain, we also evaluated the models on the remaining MMLU categories (MMLU-NoBio). As illustrated in Figure 4, aggressive interventions such as strict data filtering (Method 1) and vector negation (Method 3) induce widespread catastrophic forgetting, dragging general accuracy significantly below the baseline. While softer interventions (such as Method 2) may occasionally exhibit higher utility peaks in highly heterogeneous scenarios, they critically fail the primary safety objective. RAP-FL (Method 4) achieves the optimal trade-off between safety and retaining utility in mixed scenarios.

6 Conclusion

Our Federated Risk-Aware Projection (RAP-FL) gives the most consistent safety-utility balance among the strategies we tested, and the comparison between edge-side filtering and update-space projection is the result we regard as the most informative: in the heterogeneous scenario, keeping every sample and projecting the update out of the hazardous direction is better on both axes than discarding the hazardous samples. These findings are preliminary in a specific sense. They come from single runs of a 6.9B model on a two-client federation, and the compute this requires is what prevented us from repeating the grid over several seeds. Confirming the results either requires multi-seed runs, more clients, and a hyperparameter search.

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